

VARADIA DE MURES, Romania

WMO: 152040

Lat: 46.0192N

Lon: 22.1508E

Elev: 157

StdP: 99.45

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.7	(c) -9.0	(d) -14.7	(e) 1.1	(f) -9.6	(g) -11.5	(h) 1.4	(i) -7.4	(j) 6.0	(k) 3.2	(l) 4.4	(m) 1.1	(n) 0.5	(o) 320	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.4	(c) 33.0	(d) 21.6	(e) 31.1	(f) 21.3	(g) 29.3	(h) 20.6	(i) 23.0	(j) 29.6	(k) 22.1	(l) 29.0	(m) 21.3	(n) 28.0	(o) 1.7	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 20.9	(b) 15.9	(c) 26.0	(d) 20.1	(e) 15.0	(f) 24.4	(g) 19.3	(h) 14.3	(i) 23.5	(j) 68.5	(k) 29.6	(l) 65.4	(m) 29.1	(n) 62.5	(o) 28.1	(p) 26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.8	(b) 4.5	(c) 3.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 5.8	(b) 4.5	(c) 3.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-1.0	2.3	5.9	10.8	15.8	19.5	20.5	21.1	15.8	10.9	6.3	1.3
	DBStd	8.34	4.44	3.89	3.79	4.44	3.05	3.01	2.80	2.92	3.76	3.77	4.08	3.66
	HDD10.0	1170	341	216	135	44	2	0	0	0	2	36	125	270
	HDD18.3	3004	599	449	386	225	89	22	12	9	93	231	362	528
	CDD10.0	1465	0	0	8	69	181	284	326	344	176	64	13	0
	CDD18.3	259	0	0	0	0	9	57	79	96	17	0	0	0
	CDH23.3	3226	0	0	1	61	191	641	927	1111	255	38	1	0
	CDH26.7	1152	0	0	0	3	30	201	354	486	76	2	0	0
(14) Wind	WSAvg	1.0	0.7	0.8	1.2	1.4	1.2	1.1	1.2	1.1	0.9	0.8	0.7	0.7
(15) Precipitation	PrecAvg	727	47	41	41	58	77	85	83	54	69	63	53	60
	PrecMax	993	91	121	89	92	170	162	188	103	165	135	117	130
	PrecMin	498	15	5	11	19	33	30	26	14	8	2	1	4
	PrecStd	134	23	27	25	21	30	34	46	28	33	34	26	35
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	16.7	22.3	27.0	29.5	32.8	35.1	35.6	31.8	26.6	20.6	13.2
		MCWB	8.5	10.3	12.7	16.3	19.0	22.2	21.3	21.5	19.4	16.1	13.3	9.6
	2%	DB	8.4	13.9	19.2	25.2	27.4	31.0	32.5	33.5	28.6	24.2	17.9	9.7
		MCWB	6.0	8.9	11.1	15.4	18.8	22.2	21.5	21.1	18.8	15.7	12.3	7.6
	5%	DB	6.0	11.2	16.0	22.9	25.6	29.1	30.4	31.3	26.5	21.6	15.3	7.4
		MCWB	4.7	7.2	9.4	14.6	17.7	20.9	21.1	21.1	18.4	14.8	11.2	6.1
	10%	DB	4.4	8.7	13.4	20.3	23.7	27.4	28.3	29.3	23.4	18.3	12.6	5.9
		MCWB	3.4	6.0	8.1	13.1	16.9	20.4	20.4	20.6	17.3	13.6	10.1	4.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	11.3	13.7	18.0	21.0	24.4	23.5	24.1	21.1	17.7	14.1	10.5
		MCDB	11.2	15.5	21.6	25.2	26.7	30.6	29.6	30.9	29.6	23.3	18.8	12.6
	2%	WB	6.6	9.4	11.9	16.4	19.8	23.0	22.6	22.7	19.9	16.4	13.0	7.7
		MCDB	8.0	13.0	18.3	23.7	25.5	29.1	29.4	30.0	26.6	22.3	17.3	9.3
	5%	WB	5.0	7.9	10.3	15.2	18.6	21.9	21.9	21.8	18.9	15.4	11.8	6.4
		MCDB	5.9	10.6	15.0	22.3	23.9	27.8	28.7	29.2	24.6	20.5	14.7	7.5
	10%	WB	3.4	6.2	9.0	13.8	17.5	20.8	21.1	21.0	18.0	14.0	10.4	5.0
		MCDB	4.2	8.2	12.7	19.5	22.5	26.4	27.3	28.3	22.8	18.1	12.6	5.7
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	10.2	12.4	13.4	13.0	12.8	14.0	14.4	12.7	13.1	10.4	6.6
		MCDBR	10.2	15.1	17.4	18.9	16.9	15.8	18.0	18.5	17.9	19.3	14.7	9.4
		MCWBR	7.9	10.0	9.8	9.7	8.9	7.4	7.7	7.4	8.0	11.2	9.2	7.2
	5% WB	MCDBR	9.4	13.6	15.7	17.1	15.7	14.5	15.1	16.0	14.9	17.4	13.2	9.8
		MCWBR	7.1	9.4	9.1	8.8	8.6	7.3	7.1	7.3	7.0	10.1	8.1	7.2
(40) Clear-Sky Solar Irradiance	taub		0.332	0.360	0.401	0.457	0.443	0.436	0.452	0.459	0.426	0.385	0.355	0.332
	taud		2.339	2.267	2.205	2.064	2.165	2.227	2.189	2.164	2.224	2.333	2.370	2.368
	Ebn at Noon		752	800	816	800	828	835	814	790	781	762	718	708
	Edh at Noon		77	100	123	155	145	137	141	139	120	94	75	68
(44) All-Sky Solar Radiation	RadAvg		1.22	2.03	3.21	4.44	5.33	5.92	5.83	5.30	3.75	2.58	1.50	0.98
	RadStd		0.15	0.27	0.45	0.46	0.50	0.51	0.48	0.51	0.41	0.29	0.17	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+0.72	N/A	N/A	+0.95	N/A	N/A	-127	-190	+118	+59

Nomenclature: See separate page